MIL-P-60033B (MU)
19 April 1966
SUPERSEDING
MIL-P-60033A (MU)
29 January 1965

MILITARY SPECIFICATION

PROPELLANT, M30A1 (M30E1) FOR USE IN CHARGE, PROPELLING, 155MM, M119 (XM119E4) SUPERCHARGE (CHARGE 8) FOR HOWITZER S.P. M109

1. SCOPE

1.1 This specification covers M30A1 propellant for one type of propelling charge dsignated as 155MM Propelling Charge, M119 Supercharge, for Howitzer, S.P., M109.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids, or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-G-155 - Graphite (For Use in Ammunition)

STANDARDS

MILITARY

MIL-STD-109 - Quality Assurance Terms and Definitions
MIL-STD-652 - Propellants, Solid for Cannons, Requirements and Packing
MIL-STD-1168- Lot Numbering of Ammunition

DRAWINGS

ARMY

8882334 - Charge, Propelling, 155MM; M119 (White Bag, Charge 8) for Howtizer, S.P. M109.

FSC: 1375

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(Copies of specifications, standards, drawings and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer).

2.2 Publications.-The following document forms a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids shall apply.

CODE OF FEDERAL REGULATIONS

49 CFR 71-90 - Interstate Commerce Commission Rules and Regulations for the Transportation of Explosives and Other Dangerous Articles

(The Interstate Commerce Commission Rules are now a part of the Code of Federal Regulations (1949 Edition and revisions) available from the Superintendent of Documents, Government Printing Office, Washington, D.C. 20402. Orders for the above publication should cite "39 CFR 71-90 (latest revision)".)

3. REQUIREMENTS

3.1 Material.-The propellant shall be M30A1 (M30E1) Type I conforming to the requirements of Standard MIL-STD-652.

3.1.1 Graphite (Glaze).-The propellant grain shall be glazed with 0.1 percent maximum Graphite, MIL-G-155, Grade III or IV on added basis.

3.2 Ballistics.-The propellant, when fired in the weapon and with the weight of projectile for which it was manufactured, shall comply with the following requirements when tested as specified in 4.3.1.

3.2.1 Velocity and pressure.-At 70 degrees Fahrenheit (°F) a weight of propellant shall be carge assessed as required to comply with the requirements prescribed in Table I.

TABLE I
Velocity and Pressure

Zone	Velocity meters per second (mps)	Piezo Equivalent Pressure (pep) pounds per square inch (psi) min.	max.
8(M119 Charge	684.3	45,000	48,000

3.2.2 Uniformity of velocity.-The lot standard deviation in velocity shall be not greater than 2.00 mps.

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3.2.3 Permissable individual maximum pressure.-No individual maximum pressure shall be greater than 58,000 psi pep (115 percent of rated maximum pressure of the gun).

3.3 Workmanship.-The best commercial practices shall be used in the formulation of propellant furnished under this specification, and all other applicable documents. The propellant and its standard ingredients shall be protected from the action of direct sunlight and acid fumes. No reground propellant or nitrocellulose shall be used in the manufacture of propellant procured under this specification unless otherwise specified.

4. QUALITY ASSURANCE PROVISIONS

4.1 General quality assurance provisions.-Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements. Reference shall be made to Standard MIL-STD-109 in order to define terms used herein.

4.1.1 Submission of product.-At the time the completed lot of product is submitted to the Government for acceptance the contractor shall supply the following information accompanied by a certificate which attests that the information provided is correct and applicable to the product being submitted:

a. A statement that the lot complies with all quality assurance provisions specified within this specification.

b. Number of batches of product inspected, (shall be made available upon request by the Contracting Officer).

c. Results obtained, by defect code, for all inspection performed, (shall be made available upon request by the Contracting Officer).

d. Drawing, specification number and date, together with an identification and date of changes.

e. Certificates of conformance on all material purchased by the contractor when such material is controlled by Government or commercial specifications referenced in any of the contractual documents.

f. Number of pounds in the lot.

g. Date submitted.

h. Eight copies of propellant description sheet AMC Form 1047 (OO Form 1204) containing the applicable data (see 6.7).

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The certificate shall be signed by a responsible agent of the certifying organization. The initial certificate submitted shall be substantiated by evidence of the agent's authority to bind his principal. Substantiation of the atent's authority will not be required with subsequent certificates unless, during the course of the contract, this authority is vested in another agent of the certifying organization.

4.1.2 Government verification.-Using the contractor's written quality assurance procedure, this detail specification, the applicable drawings and other contractual documents as a guide, the Government inspector shall verify at unscheduled intervals all quality assurance operations performed by the contractor. Verification will be performed to the extent necessary to assure compliance with the contractual requirements. Severity of Government inspection of individual characteristics will be directly related to the seriousness of the classification assigned. In no instance will a characteristic classified "critical" be accepted solely on the basis of the contractor's records.

4.2 Inspection provisions

4.2.1 Lot formation.-A lot shall consist of one or more batches of M30A1 propellant produced by one manufacturer, in accordance with the same specification, same specification revision under one continuous set of operation conditions. Each lot shall consist of that quantity of M30A1 propellant which has been subjected to the same unit chemical or physical process intended to make the final product homogeneous. The criteria and procedures for the assignment of lot numbers shall be in accordance with Standard MIL-STD-1168.

5.2.2 Testing

4.2.2.1 Ballistics (see 3.2).-A ballistic test quantity of 700 pounds of the propellant shall be selected from at least 10 randomly selected containers for charge establishment, determining uniformity of velocity and determining maximum pressure. When lots are comprised of 10 containers or less, each container shall be sampled.

4.2.2.1.1 Velocity and pressure (Critical defect).-The lot of propellant shall be rejected if the average velocity of the sample rounds tested differs from the applicable required velocity in Table I by more than one percent within the pressure limits specified, or if the mean of the maximum pressures exceeds the specified maximum limit by more than one percent, or is less than the minimum limit by more than one percent in Table I.

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a. Velocity (M119 Charge)	Code No. (see 6.4 01001
b. Pressure (M119 Charge)	Code No. 02001

4.2.2.1.2 Uniformity of velocity (see 3.2.1 and 3.2.2) Major defect, Code No. 03001.-If the product of the factor 0.69 times the velocity standard deviation of the test rounds exceeds the applicable requirement, the lot shall be rejected.

4.2.2.1.3 Permissable individual maximum pressure (see 3.2.3) Critical defect, Code No. 04001.-The maximum pressure calculated for the assessed charge weight for the lot, plus four (4) times the standard deviation of the maximum pressure obtained, shall not exceed the applicable requirement (see 6.2).

4.2.2.2 Retest.-If for any reason the proof director considers that the test conditions have detrimentally affected the test results, he shall declare the results invalid and perform a new test on additional charges as required.

4.3 Testing

4.3.1 Ballistics.-Unless otherwise specified herein, the test shall be perofmred in accordance with applicable provisions of the Ordnance Proof Manual. The charge weights shall be determined by firing propelling charges assembled in accordance with applicable drawing 8882334 to meet the requirements specified in Table I. Projectile, HE, M107, (95.0 pounds) shall be used when firings are conducted utilizing the 155MM weapon.

4.3.1.1 Uniformity series.-The propellant shall be conditioned to 70°F plus or minus 2°F for at least 16 hours and shall be fired within a minimum time after removal from the conditioning chamber. A seven (7) round uniformity series shall be fired for evaluation of the production lots and the velocity mean and standard deviation shall be calculated after reducing results to standard conditions and pressure. The pressure shall be reported as piezo equivalent pressure.

4.3.2 Extreme temperature firings (see 6.6).

4.3.2.1 Conditioned rounds (minus 40 degrees F).-For informational purposes only, a seven (7) round uniformity series shall be fired on rounds which were conditioned for a minimum of 24 hours, at minus 40 degrees F. before firing. These rounds will be inert loaded except for propellant system. This test will be performed on the first 5 lots

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which have been accepted. Firings will be performed within a minimum time after removal from the conditioning chamber.

4.3.2.2 Conditioned rounds (140 degrees F).-For informational purposes only, a seven (7) round uniformity series shall be fired on rounds which were conditioned for a minimum of 15 days at 140 degrees F. before firing. These rounds will be inert loaded except for propellant system. This test will be performed on the first 5 lots which have been accepted.

4.3.3 Maximum pressure firings (see 6.8).-For informational purposes only a seven (7) round uniformity series shall be fired utilizing M121 projectiles conditioned in accordance with the provisions of 4.3.1.1. These rounds will be inert loaded except for the propellant system. This test will be performed on the first five (5) lots which have been accepted. Firings will be performed within a minimum time after removal from the conditioning chamber.

5. PREPARATION FOR DELIVERY

5.1 Packing and marking.-Packing and marking shall be in accordance with Standard MIL-STD-652.

6. NOTES

6.1 Order data.-Procurement documents should specify the following:

a. Title, number and date of this specification.
b. Packing required (see 5.1)
c. Process.-Details of the manufacturing process and the equipment used by the contractor will be submitted to the procuring activity in writing prior to commencement of manufacture.

6.2 Standard deviation.-The standard deviation shall be calculated with (N-1) as the divisor in a standard statistical technique equivalent to that shown in Standard MIL-STD-414, Section B, Example B-1.

6.3 Submission of test data.-In addition to the normal distribution of records, when the propellant is procured by the Department of Army, one copy of all ballistic test data, AMC Form 1060, will be forwarded to: Commanding Officer, Picatinny Arsenal

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ATTN: SMUPA-ND2, and one copy to ATTN: SMUPA-DC8, Dover, New Jersey.

6.4 Inspection code numbers.-The five digit code numbers assigned to the inspections herein are to facilitate future data collection and analysis by the Government.

6.5 Intended use.-It is intended that the item of this specification will be used in 155MM, M119 Supercharge, for Howitzer, S.P., M109.

6.6 Extreme temperature firings.-These tests will be performed to obtain additional information on the characteristics of the propellant at extreme temperature. Results of the informational testings will have no effect upon the acceptability of the lot(s) involved.

6.7 Submission of AMC Form 1047.-Information contained in propellant description sheet, AMC Form 1047, need not be resubmitted in the form of a certificate provided the provisions regarding certifying of responsible agents is adhered to (see 4.1.1).

6.8 Maximum pressure firings.-These tests will be performed to obtain additional information on the characteristics of the propellant when fired at maximum pressure. Results of the informational testings will have no effect upon the acceptability of the lot(s) involved.

CUSTODIAN:
ARMY-MU

PREPARING ACTIVITY:
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